

Ex. No:6 Use Sleuth Kit to analyze digital evidence

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- o Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- o and download the latest version for Windows.

1. Install Sleuth Kit:

- o Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- o Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as dd, .raw, .img, or .E01.
- o Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- o You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- o This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- o Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- o Run the command:

arduino

fsstat [image file] > filesystem_info.txt

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

arduino

mmls [image file] > partitions.txt

- This command lists the partitions within the image file.

3. Analyze File System with fls:

- Run the command:

arduino

fls -r [image file] > file_list.txt

- This command recursively lists files and directories in the file system, showing their metadata.

3. Recover Deleted Files with icat:

- To extract a specific file, use icat:

css

icat [image file] [inode number] > [output file]

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

css

istat [image file] [inode number] > metadata_info.txt

- This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

css

fls -m / -r [image file] > body.txt

- Then create the timeline:

css

mactime -b body.txt > timeline.txt

- o The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- o Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- o Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

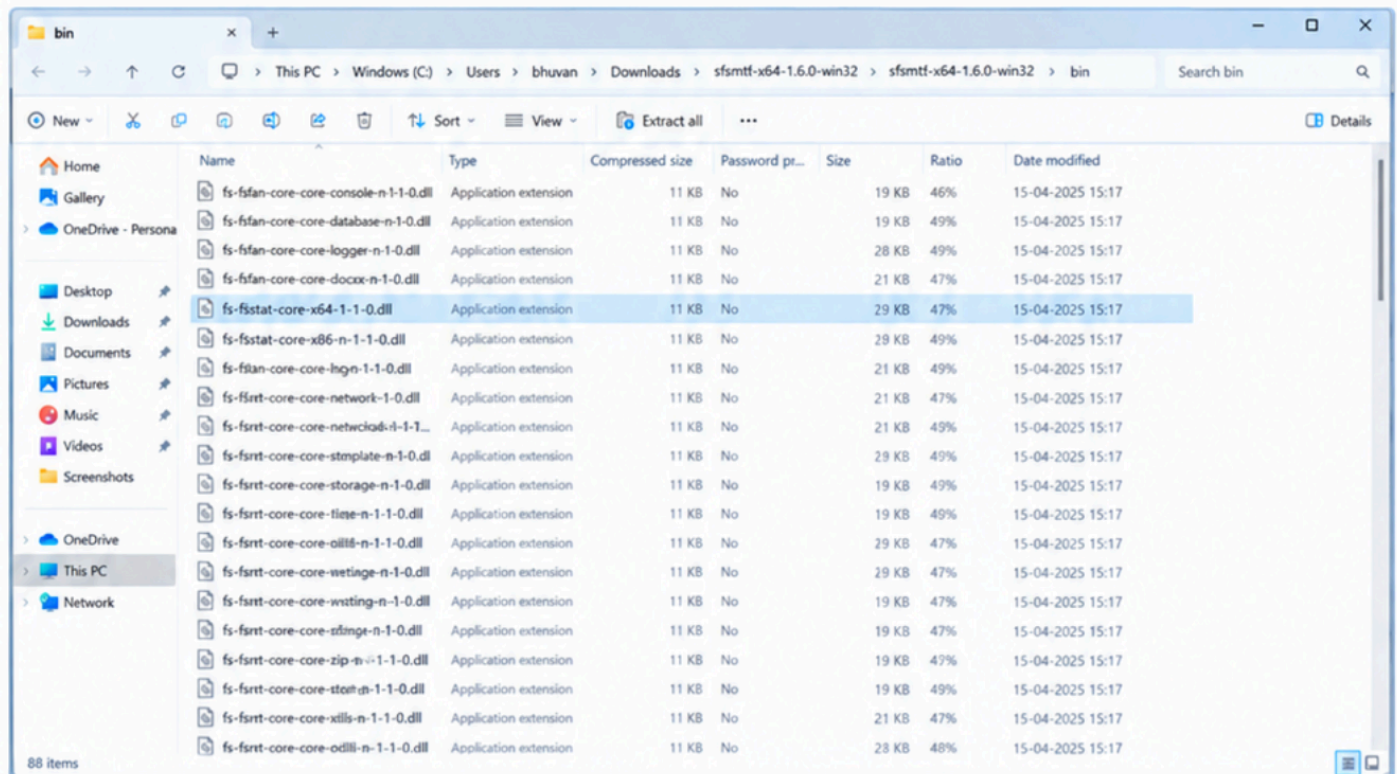
- o Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

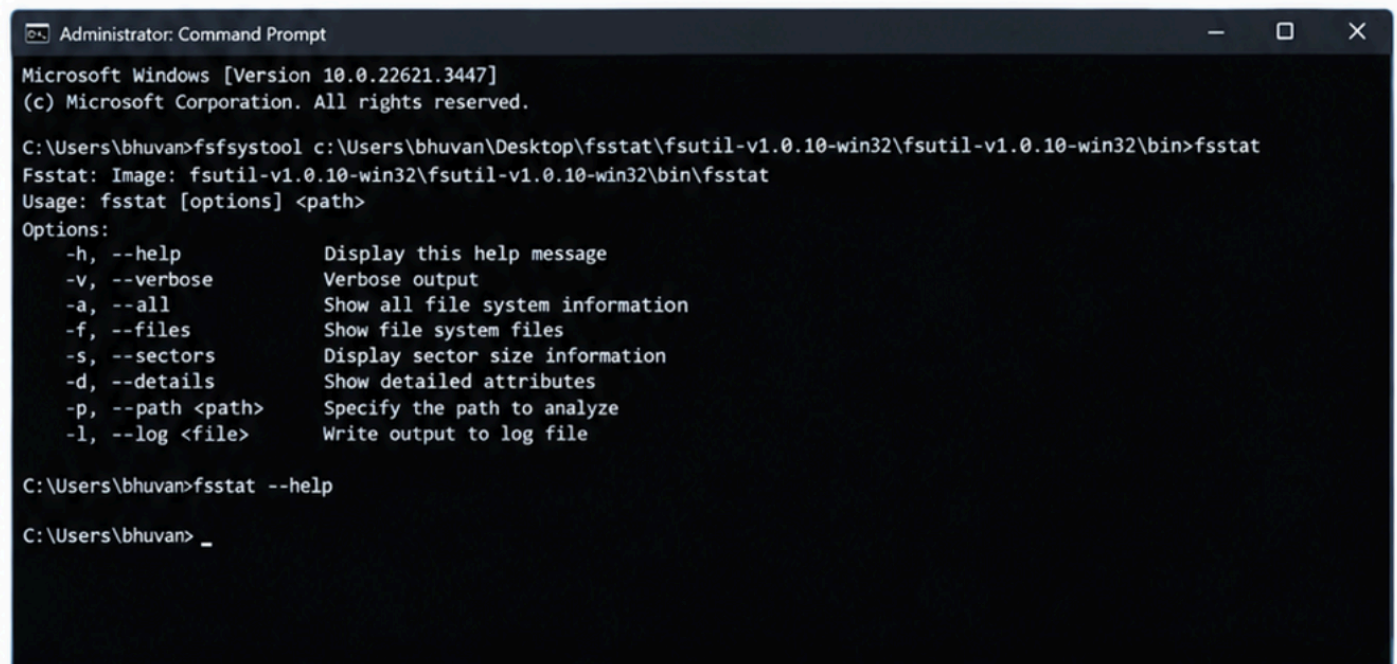
- o Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

Fsstat command execution:



Identifying File System:



Installed Sleuth Kit:

```
C:\Users\bhuvan>Reddy\Downloads\sleuthkit-4.11.0-win32\sleuthkit-4.11.0-win32\bin\img_stat ef emf -o 63 "C:\Users\bhuvan Reddy\Downloads\Welt Latitude CP1.E01"

FILE SYSTEM INFORMATION -----
File System Type: NTFS
Volume Serial Number: B8C6C1EC0B109B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION -----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4194304
Size of MFT (in bytes): 16384 bytes
Size of Index Records: 4096 bytes
Range: 2 - 131059
Root Directory: 5

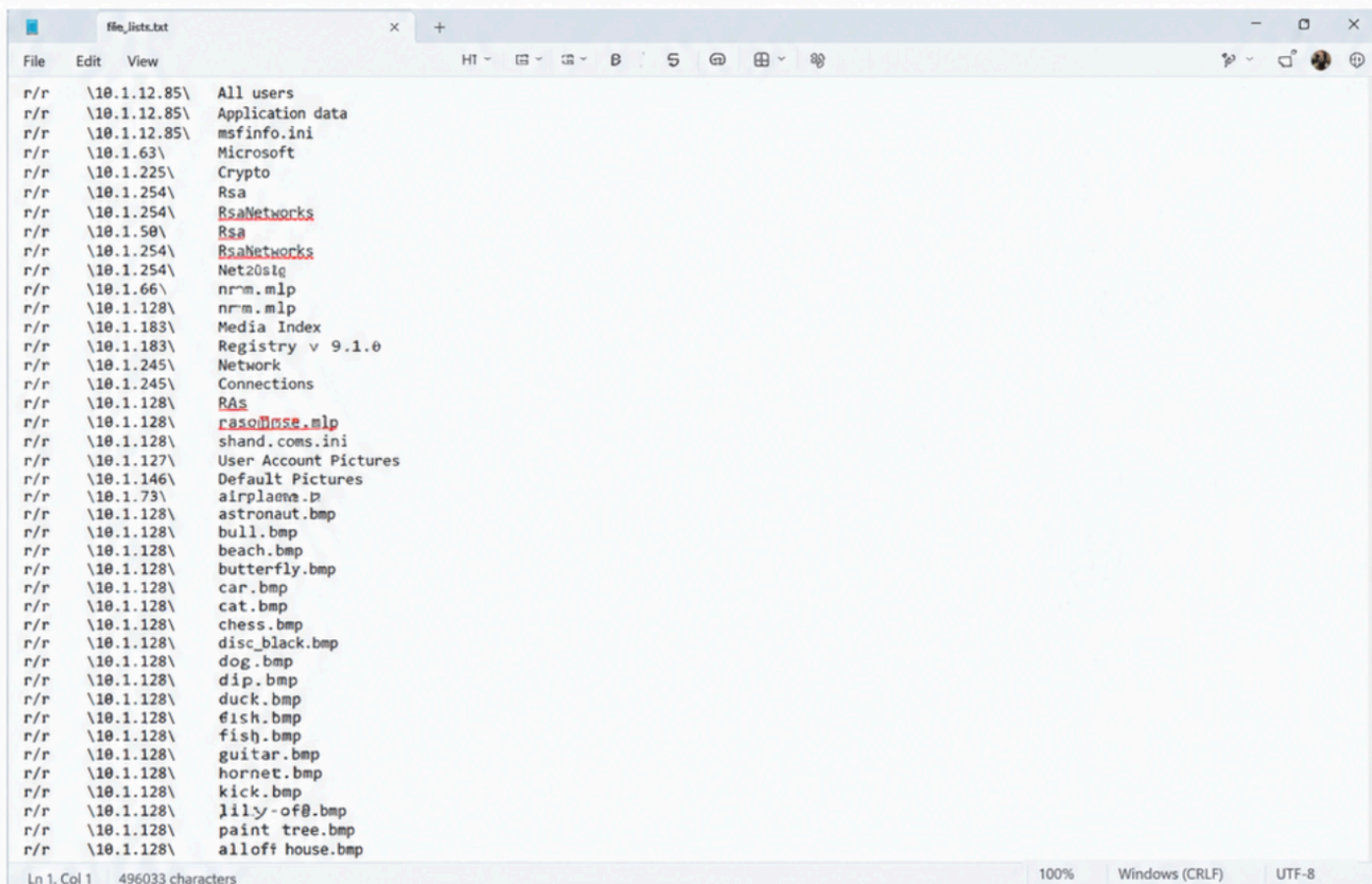
CONTENT INFORMATION -----
Sector Size: 512
Cluster Size: 512
Total Clusters Range: 0 - 8194815
Total Sectors Range: 0 - 41943015

VOLUME ATTRIBUTE VALUES:
STANDARD_INFORMATION (0x10)   Size: 48-72   Flags: Resident
ATTRIBUTE_LIST (0x20)         Size: 80-128  Flags: Non-resident
FILE_NAME (0x30)              Size: 80-128  Flags: Resident, Index
OBJECT_ID (0x40)              Size: 16-32   Flags: Resident
SECURITY_DESCRIPTOR (0x50)     Size: 72-80   Flags: Non-resident
VOLUME_NAME (0x60)            Size: 24-64   Flags: Resident
VOLUME_INFORMATION (0x70)     Size: 8-16    Flags: Resident
DATA (0x80)                   Size: 8-10    Flags: Non-resident
INDEX_ROOT (0x90)             Size: 56-120  Flags: Resident
INDEX_ALLOCATION (0xA0)        Size: 4096-8192 Flags: Non-resident
BITMAP (0xB0) (0x0000)        Size: 4096-8192 Flags: Non-resident
REPARSE_POINT (0xC0) (0x00)  Size: 16-32   Flags: Non-resident
EA_00000000 (0xD0)           Size: 8-16    Flags: Resident
EA_INFORMATION (0xE0)        Size: 36-32   Flags: Resident
EA_00000000 (0xF0)           Size: 0-0-32  Flags: Non-resident
LOGGED_UTILITY_STREAM (0x100) Size: 0-8192  Flags: Non-resident

C:\Users\bhuvan>Reddy\Downloads\sleuthkit-4.11.0-win32\sleuthkit-4.11.0-win32\bin>
```

Identifying File System:

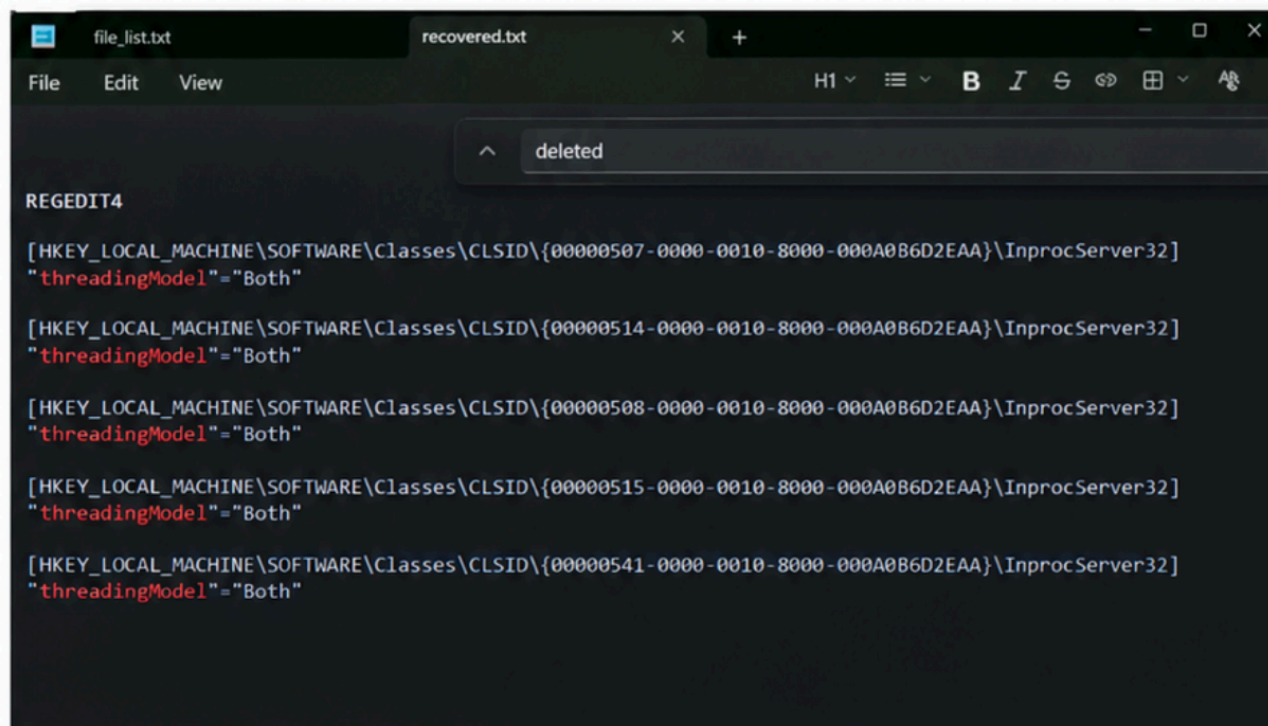
```
C:\Users\bhuvan>Reddy\Downloads\sleuthkit-4.11.0-win32\sleuthkit-4.11.0-win32\bin\fls -f emf -o 63 -r "C:\Users\bhuvan Reddy\Downloads\Welt Latitude CP1.E01"
-> file_lists.txt
```



```
file_lists.txt
File Edit View
HI - - - - - B S @ - - - - -
r/r \10.1.12.85\ All users
r/r \10.1.12.85\ Application data
r/r \10.1.12.85\ msinfo.ini
r/r \10.1.63\ Microsoft
r/r \10.1.225\ Crypto
r/r \10.1.254\ Rsa
r/r \10.1.254\ RsaNetworks
r/r \10.1.50\ Rsa
r/r \10.1.254\ RsaNetworks
r/r \10.1.254\ Net20slq
r/r \10.1.66\ nrm.mlp
r/r \10.1.128\ nrm.mlp
r/r \10.1.183\ Media Index
r/r \10.1.183\ Registry v 9.1.0
r/r \10.1.245\ Network
r/r \10.1.245\ Connections
r/r \10.1.128\ RAs
r/r \10.1.128\ raso\ase.mlp
r/r \10.1.128\ shand.coms.ini
r/r \10.1.127\ User Account Pictures
r/r \10.1.146\ Default Pictures
r/r \10.1.73\ airplane.p
r/r \10.1.128\ astronaut.bmp
r/r \10.1.128\ bull.bmp
r/r \10.1.128\ beach.bmp
r/r \10.1.128\ butterfly.bmp
r/r \10.1.128\ car.bmp
r/r \10.1.128\ cat.bmp
r/r \10.1.128\ chess.bmp
r/r \10.1.128\ disc_black.bmp
r/r \10.1.128\ dog.bmp
r/r \10.1.128\ dip.bmp
r/r \10.1.128\ duck.bmp
r/r \10.1.128\ fish.bmp
r/r \10.1.128\ fish.bmp
r/r \10.1.128\ guitar.bmp
r/r \10.1.128\ hornet.bmp
r/r \10.1.128\ kick.bmp
r/r \10.1.128\ jilly-off.bmp
r/r \10.1.128\ paint tree.bmp
r/r \10.1.128\ alloff house.bmp
Ln 1, Col 1 496033 characters 100% Windows (CRLF) UTF-8
```

Recovering Deleted File:

```
C:\Users\bhuvan>Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>icat -i ewf -o 63 "C:\Users\bhuvan\Downloads\sleuthkit-4.14.0-win32\Dell Latitude CPi (1).E01" 150 > recovered.txt
```



Metadata:

```
C:\Users\bhuvan>Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>istat -i ewf -o 63 "C:\Users\bhuvan\Downloads\sleuthkit-4.14.0-win32\Dell Latitude CPi (1).E01"
```

****FILE SYSTEM INFORMATION****

File System Type: NTFS
Volume Serial Number: BBCG61E6C08109B
OEM Name: NTFS
Version: Windows XP

****METADATA INFORMATION****

First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4194304
Size of MFT (in bytes): 16384 bytes
Size of Index Records: 4096 bytes
Range: 2 - 131089
Root directory: 5

****CONTENT INFORMATION****

Sector Size: 512
Cluster Size: 512
Total Clusters Range: 0 - 8194815
Total Sectors Range: 0 - 41943015

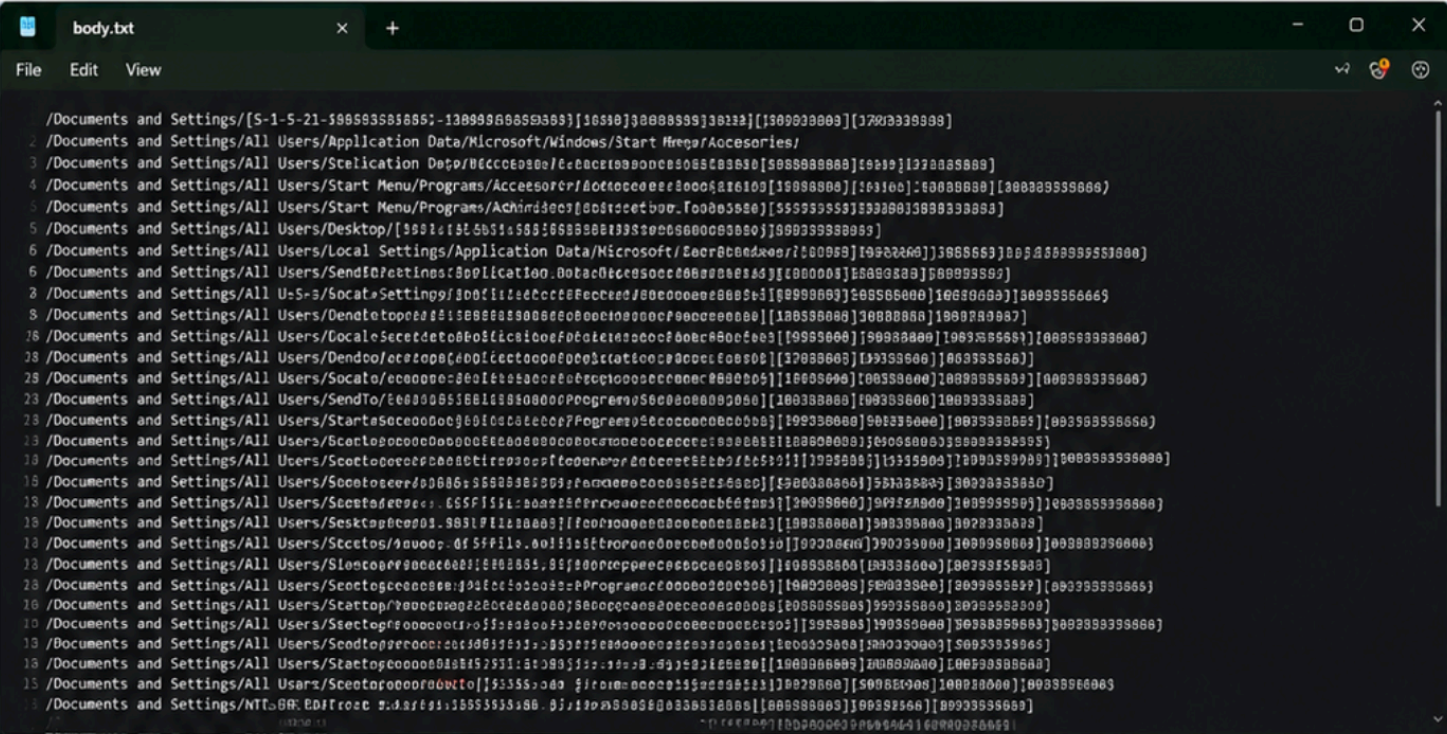
****VOLUME ATTRIBUTE VALUES****

ATTRIBUTE	Size	Flags
STANDARD_INFORMATION (0x10)	48-72	Flags: Resident
ATTRIBUTE_LIST (0x20)	88-128	Flags: Non-resident
FILE_NAME (0x30)	88-128	Flags: Resident, Index
OBJECT_ID (0x40)	16-32	Flags: Resident
SECURITY_DESCRIPTOR (0x50)	72-80	Flags: Non-resident
VOLUME_NAME (0x60)	24-64	Flags: Resident
VOLUME_INFORMATION (0x70)	8-16	Flags: Resident
DATA (0x80)	8-18	Flags: Non-resident
INDEX_ROOT (0x90)	56-128	Flags: Non-resident
INDEX_ALLOCATION (0xA0)	4096-8192	Flags: Non-resident
BITMAP (0xB0)	4096-8192	Flags: Non-resident
REPARSE_POINT (0xC0)	16-32	Flags: Non-resident
EA_INFORMATION (0xD0)	8-16	Flags: Resident
EA (0xE0)	8-32	Flags: Non-resident
LOGGED_UTILITY_STREAM (0x100)	8-8192	Flags: Non-resident

C:\Users\bhuvan>

Generating Timeline:

```
C:\Users\bhuvan\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ewf -o 63 "C:\Users\bhuvan\Downloads\sleuthkit-4.14.0-win32\bin\1"
"E01" > body.txt
```



Gathering all the generated Files(Report):

